Reservoir computer using a photorefractive nonlinear medium

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ABSTRACT

Reservoir computing (RC) is a machine learning (ML) model that has gained popularity in the recent decade due to its simplicity, versatility in solving temporal problems, and ease of implementation across various physical systems. This work explores the implementation of a photorefractive crystal as the reservoir in a reservoir computer. While previous studies have demonstrated the feasibility of using photorefractive materials for RC and other ML applications, a systematic investigation into the impact of the optical nonlinearity on the systems performance has yet to be conducted. To address this question, we develop a reservoir computer using a biased photorefractive crystal. By quantifying the system's optical nonlinearity and evaluating the performance of the reservoir computer, we reveal a correlation between these factors, providing new insights into optimizing photorefractive systems for machine learning tasks.

Keywords: Reservoir computing, Photorefractive

1. INTRODUCTION

The increasing demand for efficient and high-performance computing solutions has driven significant interest in unconventional computing paradigms. Among these, reservoir computing (RC) has emerged as a promising framework to address time-dependent problems, including signal processing, time series prediction, and pattern recognition. RC operates by utilizing a large, static recurrent neural network as a reservoir with fixed internal connections.[1] This static nature makes the reservoir directly replaceable with physical systems with sufficient dynamics such as lasers with feedback,[2,3] silicon-based ring resonators,[4] soft materials[5] or skyrmions.[6,7]

This study uses a photorefractive crystal as the reservoir, leveraging its nonlinear response to optical input induced by the photorefractive effect. Photorefractive materials were already proposed as a promising platform for ML during the late 1980s.[8] Recent works have explored various implementations of RC or extreme learning machines through simulation,[9] using different materials such as pure SBN crystals,[10] disordered $LiNbO_3$[11] or poled $LiNbO_3$.[12] Although an earlier study discussed the general physics of photorefractive crystals and their potential impact in ML applications,[13] there remains a lack of systematic investigation into the influence of optical nonlinearity on system performance, which this study aims to fill.

2. RESULTS

We use a reflective SLM to phase-encode the input on the laser before the laser passes through the crystal, and the output phase is imaged on the camera, as seen in Figure 1. The recorded image is then down-sampled before passing through a final output layer. We use an SBN:61 crystal doped with 1% Ce, subjected to an applied field along its c-axis and a halogen lamp shining on it from above for background illumination. The optical nonlinearity, or, more specifically, the refractive index modulation, will be modified by changing the applied laser power, electrical field and background illumination

The first task we use to benchmark the system's performance is the n-step XOR task. In this task, a random bit array is sequentially input into the reservoir computer, which is tasked with producing an output corresponding to the XOR operation between each bit and the bit located n steps behind in the sequence. This task serves as a straightforward measure of the system's memory capacity. In our analysis, we primarily focus

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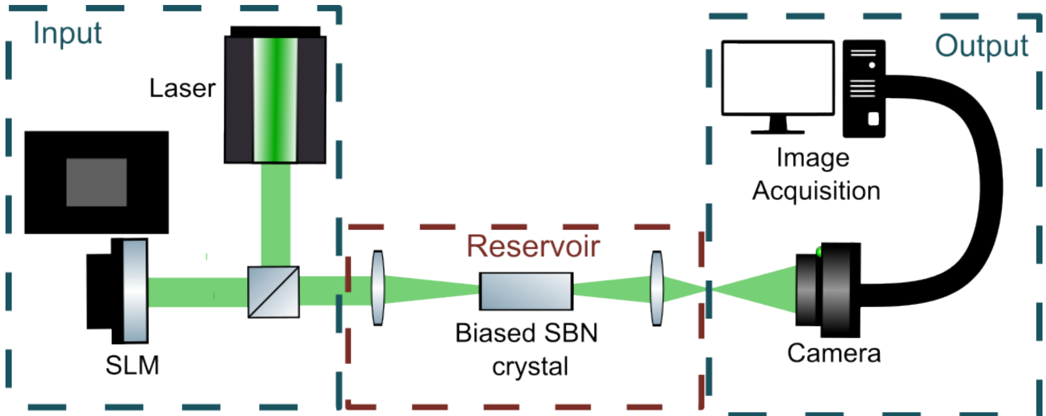

Figure 1: Simplified experimental setup. The grey and black image is an example of the input phase profile that gets encoded on the gaussian beam by the SLM.

on the n = 2 XOR, since we notice that the n = 1 XOR task achieves high accuracy even without utilizing the photorefractive crystal, due to limitations of our experimental setup.

We measure the strength of the refractive index modulation (Δn) induced by the photorefractive effect by examining the induced circular fringes in the far-field.[14] Δn increases with the applied laser intensity before saturating. The accuracy of the two-step XOR, follows a similar trend.

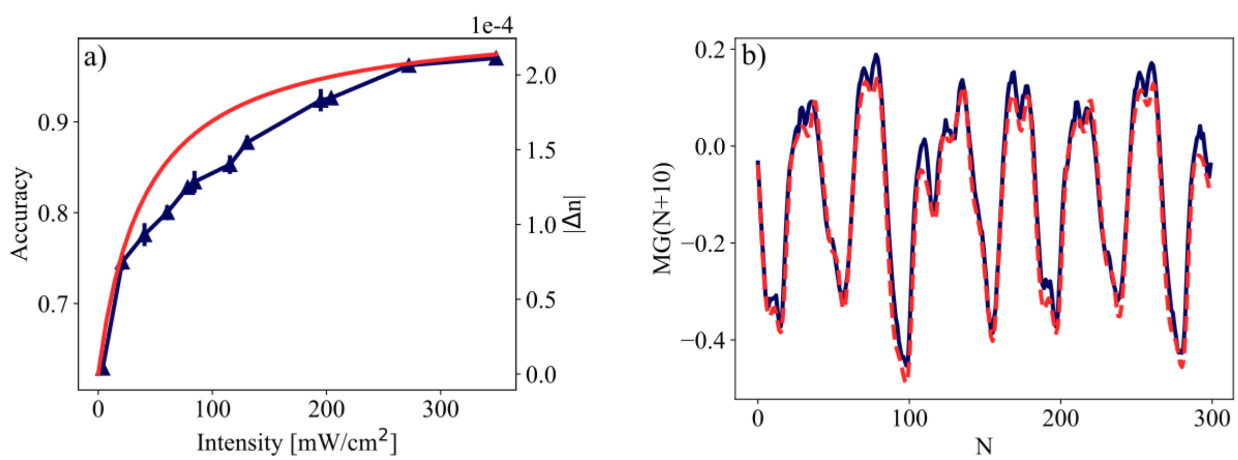

Figure 2: a) The accuracy of the 2-Step XOR task in blue. The trend of the refractive index modulation is plotted in red; the trend originates from a fit to experimental data. b) Mackey-glass chaotic time series prediction, the dashed red line represent the target.

The second task we use is Mackey-Glass chaotic time series prediction. We generate the time series similarly to that in Ref. 1 except that we use Runge-Kutta-4, with dt = 0.1, to model the dynamics. Predicting the Mackey-Glass chaotic time series requires both memory and nonlinearity in the system. After a warm-up and training phase described in Ref. 10, we task the reservoir to make an n-step prediction for every input, see Fig 2b. We use the physical parameters that reached the highest accuracy for the XOR task. The predicted data is qualitatively similar, and we use the Mean Square Error (MSE) to quantitively evaluate it, with MSE $= 5 * 10^{-3}$.

3. CONCLUSION AND OUTLOOK

We have successfully implemented a photorefractive reservoir computer and demonstrated a correlation between the refractive index modulation in our system induced by the photorefractive effect and our system's memory using the XOR task. We then used this information to successfully perform a chaotic time series prediction. In the future, we also aim to include the influence of the applied voltage and background illumination. The impact of introducing two-wave mixing into the system is also of interest. Photorefractive materials as a basis for ML offer several intrinsic advantages, a controllable NL over a large range of wavelengths, and the response times can be brought down to nanoseconds.[15]

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